

MISSOURI



MISSOURI UNIVERSITY OF SCIENCE AND TECHNOLOGY

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Section 2.1

Matrix Operations

Standard Notation

If A is an $m \times n$ matrix, then

$$A = [\mathbf{a}_1 \quad \mathbf{a}_2 \quad \cdots \quad \mathbf{a}_n]$$

where $\mathbf{a}_j$ is the vector corresponding to the j th column of A ; and

$$A = [a_{ij}] = \begin{bmatrix} a_{11} & \cdots & a_{1j} & \cdots & a_{1n} \\ \vdots & & \vdots & & \vdots \\ a_{i1} & \cdots & a_{ij} & \cdots & a_{in} \\ \vdots & & \vdots & & \vdots \\ a_{m1} & \cdots & a_{mj} & \cdots & a_{mn} \end{bmatrix}$$

where a_{ij} represents the entry in row i , column j .

Diagonal Matrices

The diagonal entries of an $m \times n$ matrix $A = [a_{ij}]$ are $a_{11}, a_{22}, a_{33}, \dots$ and together they form the main diagonal of A .

A diagonal matrix is a square matrix whose nondiagonal entries are zero.

The Zero Matrix

An $m \times n$ matrix whose entries are all zero is called a zero matrix, denoted 0 .

The size of the matrix 0 should be clear from the context, and it should also be clear from the context whether 0 refers to a scalar or a matrix.

Equality of Matrices

Two matrices are equal if they have the same size and all corresponding entries are equal.

Sums of Matrices

To add two matrices of the same size, simply add all corresponding entries.

Matrices of different sizes cannot be added together.

Scalar Multiplication

If $A = [a_{ij}]$ is an $m \times n$ matrix and c is a scalar, then

$$cA = [c(a_{ij})]$$

In other words, we simply multiply every entry of A by the scalar c .

Example 1

Let $A = \begin{bmatrix} 1 & 4 & 2 \\ 3 & 2 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 4 & 2 & 2 \\ 0 & -4 & -1 \end{bmatrix}$.
Calculate $A - 2B$.

Properties of Matrix Arithmetic

Let A , B , and C be matrices of the same size and let r and s be scalars. Then,

1. $A + B = B + A$
2. $A + B + C = (A + B) + C = A + (B + C)$
3. $A + 0 = A$
4. $r(A + B) = rA + rB$
5. $(r + s)A = rA + sA$
6. $r(sA) = (rs)A$

Matrix Multiplication

If A is an $m \times n$ matrix and B is an $n \times p$ matrix, then the product AB is the $m \times p$ matrix

$$AB = A[\mathbf{b}_1 \quad \mathbf{b}_2 \quad \cdots \quad \mathbf{b}_p] \\ = [A\mathbf{b}_1 \quad A\mathbf{b}_2 \quad \cdots \quad A\mathbf{b}_p]$$

Thus, each column of AB is a linear combination of the columns of A using weights from the corresponding column of B

Example 2

Calculate each product, if possible.

$$\begin{bmatrix} 1 & 4 & 2 \\ 3 & 2 & 0 \end{bmatrix} \begin{bmatrix} 4 & 2 & 2 \\ 0 & -4 & -1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 \\ -3 & 4 \end{bmatrix} \begin{bmatrix} 2 & -5 & 1 \\ 3 & 4 & -6 \end{bmatrix}$$

Matrix Multiplication

If AB is defined, the entry in row i and column j of AB is the sum of the products of corresponding entries from row i of A and column j of B .

Symbolically, the entries are calculated as

$$(AB)_{ij} = a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{in}b_{nj}$$

Example 3

Find the entries in the third row of the product.

$$\begin{bmatrix} 1 & -2 & 4 & 5 \\ -3 & 2 & 1 & 2 \\ 6 & 0 & 1 & 3 \end{bmatrix} \begin{bmatrix} 4 & -5 \\ 1 & 7 \\ 2 & 1 \\ -3 & 2 \end{bmatrix}$$

Properties of Matrix Multiplication

Let A , B , and C be matrices whose sizes are appropriate for the following products and let r be a scalar. Then,

1. $A(BC) = (AB)C$
2. $A(B + C) = AB + AC$
3. $(A + B)C = AC + BC$
4. $r(AB) = (rA)B = A(rB)$
5. $I_m A = A = A I_n$ if A is $m \times n$

Example 4

Let $A = \begin{bmatrix} -5 & 1 \\ 0 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 0 & 4 \\ -5 & -3 \end{bmatrix}$.

Calculate AB and BA .

Warnings About Matrix Multiplication

In general, $AB \neq BA$.

The cancellation laws do not hold for matrix multiplication. Thus, if $AB = AC$, then it is **not** true in general that $B = C$.

If $AB = 0$, we cannot conclude in general that either $A = 0$ or $B = 0$.

Powers of a Matrix

If A is an $n \times n$ matrix and k is a positive integer, then

$$A^k = \underbrace{A \cdots A}_k$$

The Transpose of a Matrix

Given an $m \times n$ matrix A , the transpose of A is the $n \times m$ matrix A^T whose columns are the corresponding rows of A .

Symbolically, if $A = [a_{ij}]$, then $A^T = [a_{ji}]$.

Example 5

Given $A = \begin{bmatrix} 1 & 4 & 2 \\ 3 & 2 & 0 \end{bmatrix}$, calculate A^T .

Properties of Matrix Transposes

Let A and B be matrices whose sizes are appropriate for the following sums and products and let r be a scalar. Then

1. $(A^T)^T = A$
2. $(A + B)^T = A^T + B^T$
3. $(rA)^T = rA^T$
4. $(AB)^T = B^T A^T$

Example 6

Let $A = \begin{bmatrix} -2 & 1 \\ 3 & 5 \end{bmatrix}$ and $\mathbf{x} = \begin{bmatrix} 4 \\ -1 \end{bmatrix}$.

Calculate each of the following quantities, if they are defined.

$$\begin{aligned} & (A\mathbf{x})^T \\ & \mathbf{x}^T A^T \\ & A^T \mathbf{x}^T \\ & \mathbf{x}\mathbf{x}^T \\ & \mathbf{x}^T \mathbf{x} \end{aligned}$$
